Supplementary Material

WORKING REVISION: PREVIOUS SUPPLEMENT RETAINED AS BASELINE

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A RL Teacher Training and Safety Details

A.1 Asymmetric Teacher Observations

Table 1 details the asymmetric actor–critic observation spaces used during teacher-policy training. The critic receives a 133-dimensional privileged simulator state, whereas the actor receives an 87-dimensional observation consisting only of signals available at deployment. We inject noise into the actor observations during training to improve robustness and reduce the sim-to-real gap. The palm and five fingertips constitute the six hand links. A "twist" is the concatenation of three-dimensional linear and angular velocity. Running mean/variance normalization is applied to both inputs and to the value targets.

Table 1: Asymmetric teacher observations.

Signal	Actor	Critic	Definition
Robot joint position q_t	13 (noisy)	13	Actuated arm–hand joint angles (rad)
Robot joint velocity $\dot{q}_t$	13 (noisy)	13	Actuated arm–hand joint velocities (rad/s)
Hand-link position	18 (noisy)	18	Palm and five fingertip positions: 6×3 Cartesian coordinates
Hand-link velocity	18 (noisy)	36	Actor: linear velocity (6×3); critic: spatial twist (6×6)
Palm contact force	–	3	Privileged Cartesian contact-force vector (N)
Measured joint torque	–	19	Privileged torque measurements for 19 instrumented joints (N m)
Object position	3 (noisy)	3	Object Cartesian position
Object orientation	4 (noisy)	4	Object rotation as a unit quaternion
Object velocity	–	6	Privileged object linear and angular velocity (twist)
Goal position	3	3	Cartesian target position for the lift task
Object scale	1	1	Scalar geometry scale supplied by the environment
Object ID	1	1	Scalar identifier for the current object
Previous action a_{t-1}	13	13	Previous 13-dimensional arm–hand command
Total dimension	**87**	**133**	

"–" denotes a signal that is unavailable to the actor. Noisy entries are sensor-like estimates; unmarked actor entries are task context or action history. Critic-only entries are used during simulation training and are not available to the deployed policy.

A.2 Teacher-Reward Formulation

We train the privileged teacher policy with a shaped per-step reward that coordinates approaching the object, forming a grasp, lifting and tracking the goal, and maintaining smooth control. At each control step t, all state and action quantities below are evaluated at that step, and the raw environment reward is the additive objective

$$\begin{aligned} r = {} & r_{\text{dist}} + r_{\text{curl}} + r_{\text{align}} + r_{\text{contact}} \\ & + r_{\text{near}} + r_{\text{goal}} + r_{\text{lift}} + r_{\text{action}}. \end{aligned} \tag{1}$$

We next define the geometric and contact quantities used by the individual sub-rewards. Let $\mathbf{p}_{o,t}$ and $\mathbf{p}_g$ denote the object and desired goal positions, respectively. For the six-element set $\mathscr{H} = \{\text{palm center}, \text{five fingertips}\}$, we define

$$d_t^{\text{hand}} = \max_{i \in \mathscr{H}} \|\mathbf{p}_{i,t} - \mathbf{p}_{o,t}\|_2, \qquad d_t^{\text{goal}} = \|\mathbf{p}_{o,t} - \mathbf{p}_g\|_2,$$
$$d_t^z = |p_{o,t}^z - p_g^z|, \qquad \Delta\mathbf{a}_t = \mathbf{a}_t - \mathbf{a}_{t-1},$$

where $\mathbf{p}_{i,t}$ is the position of hand link i, d_t^{hand} is the maximum hand-link distance to the object, d_t^{goal} is the object–goal distance, d_t^z is their vertical separation, and $\Delta\mathbf{a}_t$ is the change in action. To characterize grasp formation, for each fingertip $i \in \mathscr{F} = \{\text{index}, \text{middle}, \text{ring}, \text{little}, \text{thumb}\}$ we define

$$b_{i,t} = \mathbb{I}\left[\max_j \|\mathbf{f}_{i,j,t}\|_1 > \varepsilon_f\right], \qquad I_t^{\text{any}} = \mathbb{I}\left[\sum_{i \in \mathscr{F}} b_{i,t} > 0\right], \qquad I_t^{\text{grasp}} = b_{\text{thumb,t}}\, \mathbb{I}\left[\sum_{i \in \mathscr{F}} b_{i,t} \geq 2\right], \tag{2}$$

where $\mathbf{f}_{i,j,t}$ is one vector in the simulator contact-force matrix for fingertip i. Thus, I_t^{any} denotes any fingertip–object contact, whereas I_t^{grasp} requires the thumb and at least one other fingertip. With n_t denoting the elapsed environment steps, the eight sub-rewards in Eq. (1) are

$$r_{\text{dist}} = w_{\text{dist}} \exp(-\kappa_{\text{dist}} d_t^{\text{hand}}), \tag{3}$$

$$r_{\text{curl}} = \text{clip}\big(-w_{\text{curl}} \|\mathbf{q}_t^H - \mathbf{q}_{\text{curl}}\|_2^2, -c_{\text{curl}}, 0\big), \tag{4}$$

$$r_{\text{align}} = -w_{\text{align}} \left[\arccos\Big(\text{clip}((\mathbf{d}_t^{\text{palm}})^\top \mathbf{e}_{-z}, -1, 1)\Big)\right]^2, \tag{5}$$

$$r_{\text{contact}} = w_{\text{contact}} I_t^{\text{grasp}}, \tag{6}$$

$$r_{\text{near}} = w_{\text{near}}\, n_t\, \mathbb{I}[d_t^{\text{hand}} < \varepsilon_{\text{near}}], \tag{7}$$

$$r_{\text{goal}} = w_{\text{goal}} \exp(-\alpha_t d_t^{\text{goal}}) I_t^{\text{any}}, \tag{8}$$

$$r_{\text{lift}} = w_{\text{lift}} \exp(-\kappa_{\text{lift}} d_t^z) I_t^{\text{any}} \mathbb{I}[n_t \geq n_{\text{lift}}], \tag{9}$$

$$r_{\text{action}} = -w_{\text{action}} \left(\|\Delta\mathbf{a}_t^{\text{arm}}\|_2^2 + \lambda_H \|\Delta\mathbf{a}_t^{\text{hand}}\|_2^2\right). \tag{10}$$

Here $\mathbf{q}_t^H$ contains the actuated finger-joint positions, $\mathbf{q}_{\text{curl}}$ is the desired curling posture, $\mathbf{d}_t^{\text{palm}}$ is the world-frame unit vector along the palm-down axis, and $\mathbf{e}_{-z}$ is the world-down unit vector. The composite action is

$$\mathbf{a}_t = \begin{bmatrix} \mathbf{a}_t^{\text{arm}} \\ \mathbf{a}_t^{\text{hand}} \end{bmatrix}. \tag{11}$$

Before PPO uses the reward, the sum of all eight terms is multiplied by η_{PPO}, i.e., $r^{\text{PPO}} = \eta_{\text{PPO}} r$. The individual reward terms are not normalized separately. Table 2 summarizes the numerical parameters used in our experiments.

Table 2: Teacher-reward parameters used in all experiments.

Component	Parameter values
Hand distance	$w_{\text{dist}} = 5$, $\kappa_{\text{dist}} = 10$
Finger curl	$w_{\text{curl}} = 0.1$, $c_{\text{curl}} = 2$, $\mathbf{q}_{\text{curl}} = [0,0,0,0,0.4,0.1]$
Palm alignment	$w_{\text{align}} = 0.1$
Contact bonus	$w_{\text{contact}} = 10$, $\varepsilon_f = 10^{-4}$
Stay-close	$w_{\text{near}} = 0.005$, $\varepsilon_{\text{near}} = 0.2\,\text{m}$
Goal tracking	$w_{\text{goal}} = 8$, $\alpha_0 = 15$, $\Delta\alpha = 0.1$, $m_{\max} = 50$
Lift	$w_{\text{lift}} = 10$, $\kappa_{\text{lift}} = 8.5$, $n_{\text{lift}} = 60$ steps
Action smoothness	$w_{\text{action}} = 0.01$, $\lambda_H = 3$
Global PPO scaling	$\eta_{\text{PPO}} = 0.01$

Finally, automatic domain randomization (ADR) progressively sharpens the goal-tracking reward while leaving the lift weight fixed. We update the goal sharpness according to

$$\alpha_t = \alpha_0 + \Delta\alpha\, m_t, \qquad m_t \in \{0, \ldots, m_{\max}\}, \tag{12}$$

where m_t is the ADR increment at control step t.

A.3 Termination Conditions

We define four unsafe conditions that serve both as termination criteria during teacher-policy training and as unsafe-event signals during student-policy distillation. We additionally terminate an episode when it reaches the maximum time horizon. With $T_{\max} = 10\,\text{s}$, timeout termination was rarely observed in the reported experiments and was not classified as an unsafe event. All five termination conditions are checked at 60 Hz in the environment frame.

(a) *Harmful collision:* We monitor all arm–table contacts and trigger this termination when the palm or any fingertip is less than 1 cm above the tabletop, or when any arm link contacts the table. An arm–table contact is registered when the ℓ_1 norm of the filtered contact-force vector exceeds 0.1 N.

(b) *Object escape:* We define a permissible object workspace above the table that covers the intended manipulation region. We trigger object-escape termination as soon as the object center leaves this space or falls below the table.

(c) *Palm flipped:* The palm should face downward toward the object and table. We trigger this termination when its downward direction differs from world-down by more than 90°, at which point the nominal grasp geometry is inverted.

(d) *Hand too far:* We use the same permissible workspace as in (b) and trigger this termination when the palm link leaves it, indicating that the hand has moved too far during exploration.

(e) *Timeout:* We terminate the episode when the time index reaches the maximum episode horizon, $t \geq T_{\max}$.

Expanding the termination rule used in the main paper gives

$$\mathbb{I}_{\text{term}}(s_t, t) = \mathbb{I}_{\text{coll}}(s_t) \vee \mathbb{I}_{\text{obj}}(s_t) \vee \mathbb{I}_{\text{flip}}(s_t) \vee \mathbb{I}_{\text{hf}}(s_t) \vee \mathbb{I}_{\text{timeout}}.$$

To avoid conflicts between simultaneous termination conditions, we assign the event label using the following descending severity priority: harmful collision $\succ$ object escape $\succ$ palm flipped $\succ$ hand too far $\succ$ timeout.

B VLM Prompt and Inference Details

B.1 Visual-Input Storage and Example

Here, we provide representative examples of the selected visual inputs in Fig. 1. These images illustrate observations associated with the two intervention sources considered by our framework: high teacher–student disagreement and low predicted task success. The algorithm block that follows presents the semantic prompt, which combines these visual samples with recent rollout statistics and instructs the VLM to recommend conservative updates to the corresponding intervention thresholds.



(a) (b) (c) (d)

Figure 1: Representative visual inputs selected according to the two intervention signals. Samples (a)–(c) were selected for high teacher–student disagreement, with ℓ_2 values 9.7506, 17.0997, and 51.9710, respectively; (d) was selected for low predicted success (0.2734). All four records have `unsafe=true`.

The algorithm block gives the full prompt template used by the runtime implementation. Braced terms denote values serialized at query time, and the visual-sample block is repeated once for each attached image.

VLM Prompt Example

```
System message:
You tune safety-threshold displacements and return
compact JSON only.
User message:
You are a conservative threshold advisor for
DexSafeDagger robot distillation.
Do not decide intervention directly. Recommend threshold
displacements for the existing gates.
Task: Propose displacements for the teacher-student
disagreement and predicted-success thresholds used in
dexterous robot distillation.
Existing arbitration:
- intervene if teacher_student_l2 >= l2_threshold
- or if predictor_success <= success_threshold
Return only strict JSON with keys: delta_l2_threshold,
delta_success_threshold, confidence, reason.
Keep changes conservative. Prefer small threshold
shifts unless the recent window shows clear over- or
under-intervention.
Base thresholds and clamps:
- base_l2_threshold={base_l2_threshold}
- allowed_l2_range=[{l2_min}, {l2_max}]
- base_success_threshold={base_success_threshold}
- allowed_success_range=[{success_min}, {success_max}]
Current advisor thresholds:
- current_l2_threshold={current_l2_threshold}
- current_success_threshold={current_success_threshold}
Recent rollout statistics:
{
 "frame": {frame}, "intervention_rate": {beta}, "step":
{step},
 "sample_count": {N}, "l2_threshold": {tau_delta},
 "success_threshold": {tau_Q}, "unsafe_episode_rate":
{r_unsafe},
 "l2": {"min": {v}, "max": {v}, "mean": {v},
      "p10": {v}, "p50": {v}, "p90": {v}},
 "success": {"min": {v}, "max": {v}, "mean": {v},
      "p10": {v}, "p50": {v}, "p90": {v}},
 "unsafe_reason_prop": { "hand_too_far": {v},
      "harmful_collision": {v}, "object_out_of_bound":
{v},
      "palm_flipped": {v}},
 "visual_buffer_size": {B}, "visual_samples_attached":
{M},
 "visual_samples": [{ "env_id": {v}, "frame": {v},
      "image_attached": true, "image_key": {v},
"source": {high_l2 | low_success | unsafe_triggered},

      "l2_threshold": {v}, "success_threshold": {v},
      "teacher_student_l2": {v}, "predictor_success":
{v},
      "object_id": {v}, "object_name": {v},
"object_env_count": {v},
      "step": {v}, "unsafe": {true | false}}, ...]
}
For each selected visual sample i = 1,...,M:
Visual sample {i} metadata: {the corresponding record
above, excluding "image_attached"}
<IMAGE {i}: JPEG data URL, detail="high">
```

B.2 Model and Input Configuration

We used OpenAI GPT-5.5 with the rollout statistics above and up to eight high-detail RGB images encoded as JPEG data URLs. We queried the model after 2,000 episodes and every 10,000 episodes thereafter (10 calls over 100,000 episodes). Archived reports from the earlier low-detail setting contain 30,641 prompt and 4,841 completion tokens (35,482 total).

B.3 Output Processing and Threshold Updates

The requested machine-readable output is a top-level JSON object with the following schema:

```
{
  "delta_l2_threshold": <number>, "delta_success_threshold": <number>,
  "confidence": <number>, "reason": <string>
}
```

The two returned values are threshold displacements. For each channel $x \in \{\delta, Q\}$, the candidate threshold is

$$\tilde{\tau}_x^{(k)} = \tau_x^{(k)} + \Delta\tau_x^{(k)}.$$

Each candidate is clipped to a fixed range around its initial value and then exponentially smoothed with the previously applied threshold:

$$\bar{\tau}_x^{(k)} = \mathrm{clip}\Big(\tilde{\tau}_x^{(k)}, c_x^{\min}\tau_x^{(0)}, c_x^{\max}\tau_x^{(0)}\Big), \qquad \tau_x^{(k+1)} = (1-\alpha)\tau_x^{(k)} + \alpha\bar{\tau}_x^{(k)}.$$

We use $\alpha = 0.1$ and $(c_x^{\min}, c_x^{\max}) = (0.5, 1.5)$ for both thresholds. The updated values are applied from the next control step through the intervention gate $\|a_t^S - a_t^T\|_2 > \tau_\delta \ \vee \hat{Q}_{\mathrm{succ}}(s_t, a_t^S) < \tau_Q$; lowering τ_δ or raising τ_Q increases teacher intervention. The `confidence` and `reason` fields are logged for diagnostics and do not alter the update. This adaptation is used only during simulation distillation; physical deployment uses the frozen RGB student alone.

If a VLM request fails or its output cannot be parsed, the controller retains the previously applied thresholds and tries again at the next scheduled update. None of the reported trials encountered a condition that activated this fallback.

C Additional Experimental Results

C.1 Bellman Critic Evaluation

We use the resulting student and fixed-threshold Bellman critic from ablation (c) to evaluate the critic's performance. Both networks are frozen during evaluation, and the critic runs in parallel as a passive evaluator. We evaluate 200 episodes for each of the 40 training objects in simulation. Table 3 reports the observed lift-success and unsafe-episode rates together with the task-outcome prediction accuracy, the unsafe-episode detection rate, and its corresponding false-negative rate.

The lift-success rate and unsafe-episode rate characterize the student policy's ability to accomplish the task and the frequency of unsafe outcomes, respectively. Task-outcome prediction accuracy is the percentage of episodes for which the critic correctly predicts the task outcome. The unsafe-episode detection rate is the percentage of unsafely terminated episodes that are flagged by the critic before termination. The unsafe false-negative rate is the complementary percentage of unsafe episodes that the critic does not flag before termination.

Table 3: Evaluation of the frozen student and fixed-threshold Bellman critic obtained from ablation (c) under student-controlled rollouts.

Metric	Result
Lift-success rate	86.8
Unsafe-episode rate	11.9
Task-outcome prediction accuracy	90.3
Unsafe-episode detection rate	94.2
Unsafe false-negative rate	5.8

C.2 Ablations Across Random Seeds

We independently train four intervention variants—Vanilla DAgger (a), Vanilla SafeDAgger (b), critic-based intervention (c), and VLM-guided intervention (d)—using five random seeds, $\{40, 41, 42, 43, 44\}$. For each variant and seed, we freeze the resulting student and evaluate it for 200 episodes on each of the 40 training objects in simulation, yielding 8,000 evaluation episodes per seed. Because all variants use the same number of training samples and the teacher-intervention rates exhibit only minor variation across seeds, Table 4 reports only the final lift-success and unsafe-episode rates as the mean $\pm$ standard deviation across the five seeds.

Table 4: Comparison of Vanilla DAgger, Vanilla SafeDAgger, critic-based intervention, and VLM-guided intervention across five random seeds $\{40, 41, 42, 43, 44\}$. Lift success and unsafe rate are reported as mean $\pm$ standard deviation across seeds.

Metric	Vanilla DAgger (a)	Vanilla SafeDAgger (b)	Critic-based intervention (c)	VLM-guided intervention (d)
Lift Success (%) $\uparrow$	87.1 ± 0.99	86.3 ± 0.92	86.8 ± 0.74	87.5 ± 0.98
Unsafe Rate (%) $\downarrow$	17.4 ± 0.52	16.7 ± 0.48	11.9 ± 0.58	9.3 ± 0.39

C.3 Complete Real-World Evaluation and Demonstrations



Figure 2: **Sixteen-object demonstration inventory:** (1) Horse, (2) Teapot, (3) Sphinx, (4) Dragon, (5) Homer bust, (6) Ogre, (7) Cheburashka, (8) Lucy statue, (9) Beethoven bust, (10) Igea bust, (11) Dog pair, (12) Cow statue, (13) Buddha statue, (14) Blub character, (15) Monkey head, and (16) Dinosaur.

We report results for the remaining 16 real-world test objects that are not shown in the manuscript's six-object table, using the same success and safety breakdown. The two eight-object blocks below separate task success from the observed unsafe failure modes.

Table 5: Per-object results for the remaining 16 objects (30 episodes per object). **OOD objects** are shaded.

Metric (%)	Horse	Teapot	Sphinx	Dragon	Homer	Ogre	Chebura shka	Lucy
Lift Success	83.3	90.0	80.0	86.7	76.7	83.3	90.0	80.0
Unsafe Rate Avg	16.7	6.7	13.3	13.3	23.3	10.0	6.7	16.7
Harmful collision	6.7	3.3	3.3	3.3	3.3	3.3	3.3	3.3
Object escape	6.7	3.3	6.7	10.0	10.0	6.7	3.3	10.0
Palm flipped	3.3	0.0	3.3	0.0	6.7	0.0	0.0	3.3
Hand too far	0.0	0.0	0.0	0.0	3.3	0.0	0.0	0.0

Metric (%)	Beethoven	Igea	**Dog Pair**	**Cow statue**	**Buddha statue**	**Blub**	**Monkey head**	**Dinosaur**
Lift Success	86.7	73.3	**70.0**	**63.3**	**73.3**	**60.0**	**66.7**	**56.7**
Unsafe Rate Avg	6.7	13.3	**33.3**	**40.0**	**33.3**	**43.3**	**36.7**	**46.7**
Harmful collision	0.0	3.3	**13.3**	**10.0**	**13.3**	**23.3**	**16.7**	**26.7**
Object escape	6.7	3.3	**13.3**	**16.7**	**13.3**	**16.7**	**16.7**	**16.7**
Palm flipped	0.0	3.3	**3.3**	**10.0**	**3.3**	**3.3**	**3.3**	**3.3**
Hand too far	0.0	3.3	**3.3**	**3.3**	**3.3**	**0.0**	**0.0**	**0.0**

Table 6 summarizes performance across all 22 objects. The mean rates reproduce the pooled results in Main Paper Table IV; the 95% confidence intervals account for the 420 seen-object episodes and 240 unseen-object episodes.

Table 6: Aggregate real-world performance across 14 seen and 8 unseen objects. Mean rates are pooled over 30 episodes per object (420 seen and 240 unseen episodes); parentheses report Wilson 95% confidence intervals.

Metric (%)	Seen mean (95% CI)	Unseen mean (95% CI)
Lift success	83.6 (79.7–86.8)	64.6 (58.3–70.4)
Unsafe rate avg.	14.7 (11.7–18.5)	39.2 (33.2–45.5)
Harmful collision	4.7 (3.1–7.2)	18.8 (14.3–24.2)
Object escape	8.1 (5.9–11.1)	15.8 (11.8–21.0)
Palm flipped	1.4 (0.7–3.1)	3.3 (1.7–6.4)
Hand too far	0.5 (0.1–1.7)	1.3 (0.4–3.6)

C.4 Physical Baseline Comparison

We deploy the frozen RGB student policies resulting from ablations (c) and (d) on the real robot. Ablation (c) uses our RL teacher-training recipe with the stationary DexSafeDagger intervention policy, whereas ablation (d) uses the same teacher recipe with VLM-guided threshold adaptation. We evaluate both policies using the same objects, initial-condition protocol, and number of trials. We report lift-success and unsafe-event rates separately for the 14 seen and 8 unseen objects in Table 7.

Table 7: Overall real-world comparison of the frozen student policies resulting from ablations (c) and (d). Rates are reported separately for the 14 seen and 8 unseen objects.

Metric (%)	Ablation (c)		Ablation (d)	
	Seen (14)	Unseen (8)	Seen (14)	Unseen (8)
Lift success	83.4	61.7	83.6	64.6
Unsafe-event rate	16.1	41.4	14.8	39.2

The comparison supports our claim that the student trained with VLM-augmented intervention achieves a comparable real-world task-success rate while exhibiting a lower unsafe-event rate for both seen and unseen objects.

C.5 Implementation Details of Ablation (f)

Ablation (f) replaces the VLM tuner in ablation (d) with a deterministic performance-triggered threshold curriculum inspired by the budget-aware intervention principle of ThriftyDAgger. The teacher, student, Bellman critic, intervention gate, admissible threshold range, rollout budget, and update schedule are otherwise identical to those in ablation (d); the curriculum uses neither language nor visual inputs.

At scheduled update k, let $\mathscr{W}_k$ denote the most recent rollout window. We compute its lift-success and unsafe-episode rates as

$$\bar{s}_k = \frac{1}{|\mathscr{W}_k|} \sum_{\tau \in \mathscr{W}_k} \mathbb{I}_{\text{succ}}(\tau), \qquad \bar{u}_k = \frac{1}{|\mathscr{W}_k|} \sum_{\tau \in \mathscr{W}_k} \mathbb{I}_{\text{unsafe}}(\tau). \tag{13}$$

The curriculum contains five ordered operating points, indexed from the most conservative ($j = 0$) to the most autonomous ($j = 4$):

j	0	1	2	3	4
$c_\delta^{(j)}$	0.50	0.75	1.00	1.25	1.50
$c_Q^{(j)}$	1.50	1.25	1.00	0.75	0.50

(14)

At level j_k, the intervention thresholds are

$$\tau_\delta^{(k)} = c_\delta^{(j_k)} \tau_\delta^{(0)}, \qquad \tau_Q^{(k)} = c_Q^{(j_k)} \tau_Q^{(0)}. \tag{15}$$

Thus, moving toward a smaller level lowers the disagreement threshold and raises the minimum acceptable predicted-success threshold, producing more teacher intervention. Moving toward a larger level has the opposite effect and grants the student more autonomy. The resulting gate remains an OR rule: intervention occurs when either $\|a_t^S - a_t^T\|_2 > \tau_\delta^{(k)}$ or $\hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q^{(k)}$.
Let $u_{\max}$ be a validation-selected unsafe-rate limit and $s_{\min}$ a validation-selected competence threshold. Define $A_k = \mathbb{I}[\bar{u}_k \leq u_{\max} \wedge \bar{s}_k \geq s_{\min}]$. Starting from the nominal operating point $j_0 = 2$, the curriculum update is

$$j_{k+1} = \begin{cases} \max(j_k - 1, 0), & \bar{u}_k > u_{\max}, \\ \min(j_k + 1, 4), & A_k = 1 \ \wedge \ A_{k-1} = 1, \\ j_k, & \text{otherwise.} \end{cases} \tag{16}$$

Consequently, one unsafe window immediately moves the gate one level toward more teacher intervention, whereas two consecutive windows satisfying both safety and task-performance criteria are required before increasing student autonomy. This asymmetric update prevents rapid oscillation. We apply the first curriculum update after 2,000 distillation episodes and subsequent updates every 10,000 episodes, matching the VLM update schedule in ablation (d).

D Additional Material for the Current Revision

This section extends the previously submitted supplementary material in direct response to the latest review round. All red bracketed text is an internal drafting prompt and must be replaced with verified material before submission.

D.1 Teacher-Training Recipe Scope

The submitted DexTrAH/FABRICS baseline was adapted and retuned for the Tiangong 2 Pro and InspireHand, including the robot geometry, constraints, action interface, and controller parameters. It retained DexTrAH's human-motion-derived PCA hand-action manifold and the FABRICS controller between the RL policy and the robot joints. Our retained teacher implementation instead follows the normalized direct motor-command formulation used in UniDexGrasp++ (Wan et al., ICCV 2023): it removes FABRICS and maps 13 normalized policy outputs one-to-one to joint-position targets for the seven arm and six hand DoFs. In our platform-specific comparison, this direct-control recipe achieved at least comparable task performance while avoiding human-motion retargeting, PCA action-manifold construction, and embodiment-specific FABRICS tuning. We treat this as a practical implementation advantage rather than evidence that the underlying RL algorithm is intrinsically superior. The two FABRICS-based ablations were therefore removed from the primary contribution argument. All retained ablations use the same teacher setup and compare only the student-distillation and intervention mechanisms.

[Insert the verified teacher-only comparison. Panel (a): evaluation lift-success versus environment steps for the retuned DexTrAH/FABRICS and direct-control teachers, averaged over matched seeds with 95% confidence intervals. Panel (b): paired final lift-success differences over all 40 objects with the pre-specified equivalence band. Panel (c): teacher unsafe-termination rate versus environment steps.]

Figure 3: Planned teacher-only comparison on the Tiangong 2 Pro–InspireHand platform. Both implementations must use the same objects, initial-state distribution, episode horizon, evaluation episodes, PPO interaction budget, and random seeds. This figure will document the retuned DexTrAH/FABRICS transfer and test the scoped claim of comparable teacher performance without involving student distillation.

D.2 Critic False Negatives, Lead Time, and Threshold Tradeoff

[State whether each critic metric is computed per episode or per time step. Report the verified false-negative numerator and denominator, break missed warnings down by unsafe-event type, and discuss the consequence of a missed critic warning in the two-channel OR intervention gate.]

For every unsafe episode, define t_{warning} as the first control step at which the critic crosses the intervention threshold and t_{unsafe} as the first unsafe-event step. At a 60 Hz control rate, the warning lead time is

$$\Delta t_{\text{lead}} = \frac{t_{\text{unsafe}} - t_{\text{warning}}}{60} \text{ s}.$$

[Report the median, interquartile range, and 10th–90th percentile of lead time, together with the fractions warned at least 0.1 s, 0.25 s, and 0.5 s before termination. If time-step scores were not saved, state that this analysis cannot be reconstructed. If saved scores permit, add a critic-threshold sweep covering false-negative rate, intervention rate, lift success, and unsafe-event rate.]

D.3 Real-Robot Statistical Comparison

[State whether the physical trials for ablations (c) and (d) are paired by object, seed, initial pose, and other initial conditions. Report effect sizes, confidence intervals, the justified paired or unpaired test, the test statistic, and p-value separately for seen and unseen objects. State plainly whether each difference is statistically distinguishable.]

D.4 Final VLM Configuration, Latency, Cost, and Scope

[Audit and update the earlier VLM section with the exact final model identifier and provider/API version, prompt, decoding parameters, image detail and resolution, visual-sample selection rule, output schema, parsing and clipping logic, update interval, failure fallback, measured latency distribution, token counts, and monetary cost. Distinguish measured values from estimates and remove provisional or unverifiable model names.]

[State exactly which evaluation factors varied, including object geometry, texture, material, lighting, background, camera, and hazard semantics. State explicitly whether novel scene semantics were evaluated; if not, limit the claim to the tested variation.]

D.5 Unsafe-Event Observability

Table 8: Observability of the task-specific unsafe-event definitions.

Event	Simulation signal	Hardware-observable signal	Evaluation label source
Harmful collision	**[signal]**	**[signal]**	**[source]**
Object escape	**[signal]**	**[signal]**	**[source]**
Palm flipped	**[signal]**	**[signal]**	**[source]**
Hand too far	**[signal]**	**[signal]**	**[source]**

D.6 Aggregation Populations and Displayed-Object Selection

[Define the population and aggregation rule for every reported mean. Document the true selection procedure for the four displayed simulation objects and the two Figure 4 objects, including whether each selection was fixed before inspecting comparative results. Add a print-quality version of Figure 4 or complete per-object failure-mode results if available.]

D.7 Additional Limitations

[Extend the existing limitations discussion to cover the approximately 39% unsafe-event rate on unseen real objects, sim-to-real contact-strategy mismatch, dependence on an external VLM, task-specific unsafe-event definitions, hardware-versus-simulation observability differences, and the absence of formal constraint guarantees.]